ASN unit-3 - Adhoc sensor networks notes

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Unit-3

Geocasting:-

→ Geocasting is a networking technique used to deliver data to a specific geographical region rather than to a specific device or node.

→ It Plays a critical role in applications like disaster recovery, military operations, and environmental monitoring, where efficient and location-based communication is required.

Data-transmission Oriented-LBM:-

→ location-Based multicast is a routing technique that uses geographical information to deliver data from a source node to multiple destination nodes within a specific area.

→ In a data transmission-oriented LBM, the Primary focus is optimizing the transmission of data for Improved efficiency, reduced latency and lower resource Consumption

* key features:-

1. Geographical Awareness:-
→ Nodes use location information to determine their Positions relative to the multicast region.

2. Multicast Region Definition:-
→ The multicast region is defined by a geometric area rather than a list of individual recipient nodes.

3. Data Transmission optimization:-
→ Efficient routing and forwarding mechanisms are designed to reduce energy Consumption, latency, and redundant transmissions.

4. Dynamic Adaptability
→ Designed to handle dynamic changes in n/w topology, which is Common in mobile ad hoc networks and WSN.

* How Data Transmission-oriented LBM works:-

1. Region Specification:-
→ The Source node specifies a geographic region for the multicast.
→ The message includes the region's boundaries, often in the form of latitude and longitude coordinates.

2. Packet forwarding:-
→ Nodes outside the region ignore the message, while those inside or near the region forward it.
→ Position-based routing protocols like Greedy Perimeter Stateless Routing (GPSR) or geographic forwarding are commonly used.

3. Data Distribution:-
→ once Packets reach the region, they are disseminated to all nodes within the area using strategies like flooding, hierarchical multicast or energy-efficient forwarding.

4. Acknowledgment:-
→ In some applications, recipient nodes may send acknowledgments to confirm data receipt, through this adds to overhead.

* Advantages:-
1. Scalability
2. Energy Efficiency
3. Reduced latency
4. Adaptability

* Challenges:-
1. Localization Errors
2. Network Density
3. Interference and Collisions
4. Security

* Applications:-
1. Disaster recovery
2. Environmental monitoring
3. Vehicular Ad hoc Networks (VANETs)
4. Military and Tactical Networks.

* Protocols used:-
1. Geocast Routing Protocol
2. Cluster-Based Multicast Protocol
3. Energy-Efficient Multicast Protocol.

Route creation oriented-GeoTORA.
→ GEOTORA stands for Geographic Temporally ordered Routing Algorithm.
→ GeoTORA is a routing Protocol specifically designed for geocasting in MANETS and WSN.

→ Geocasting involves delivering data to a group of nodes within a specific geographic region.

→ It leverages the geographical Positions of nodes to optimiz route creation, reduce overhead, and enhance scalability in dynamic n/w environments.

* <u>Assumptions</u>:-

1. Node Positioning:-

→ Nodes know their geographic locations via GPS or other localization techniques.

2. Position-based forwarding:-

→ Routing decisions rely on the relative Positions of nodes rather than traditional routing tables.

3. Topology Adaptability:-

→ The algorithm dynamically adapts to changes in n/w topology, such as node mobility or failure.

* Route <u>Creation in Geo</u>TORA:-

1. Initialization:-

→ Each node discovers its neighbors and their geographic location.

→ Nodes within the geocast region are identified.

2. Route Establishment:-

→ A source node initiates the route Creation process.

→ It broadcasts a route request message Containing the geographic coordinates of the desired region.

→ Nodes within the region respond with their location information.

3. Route Construction:-

→ The Source node Constructs a directed acyclic graph (DAG) based on the received location information.

→ The DAG represent the optimal Paths to reach nodes within the geocast region.

4. Route maintenance:-

→ Periodically, nodes Exchange routing information to update the DAG.

→ This ensures that the routes remain valid as nodes move or n/w Conditions Change.

* optimizations in GEOTORA:-

1. Energy Efficiency:-

→ GEOTORA minimizes energy Consumption by reducing redundant transmissions.

2. Scalability:-

→ Geographic routing avoids maintaining Complex routing tables, making it suitable for large-scale n/w.

3. Robustness:-

→ The algorithm Supports topology changes and node mobility without significant Performance degradation.

* Applications:-

1. Wireless sensor networks (WSNs)
2. Vehicular Ad hoc Networks (VANETs)
3. Mobile Ad Hoc Networks (MANETs)

* challenges:-

1. Scalability
2. Mobility
3. Energy Consumption.

MGR:-

→ MGR stands for Minimum Geographic Routing.

→ MGR is a geographic-based routing protocol designed for efficient Communication in ASN.

→ It minimizes the routing cost by leveraging the Physical locations of nodes and selecting Paths that reduce overall transmission distance or energy consumption.

* key features of MGR

1. Position-Based forwarding:-

→ MGR relies on node location information.

→ Each node knows its Position and that of the destination.

2. Greedy Routing:-

→ Routes are determined by forwarding data Packets to the neighbor closest to the destination.

→ This ensures a minimum geographic distance is Covered at each hop.

3.

3. Adaptability:-
→ MGR is suitable for dynamic environments, adjusting to node mobility or failure.
→ It operates without the need for pre-established routing tables.

4. Energy Efficiency:-
→ By minimizing transmission distance, MGR conserves energy, a critical factor for resource-constrained devices in sensor n/w.

* How MGR works:-

1. Node Discovery:-
→ Each node discovers its neighbors and their geographic locations.

2. Packet Reception:-
→ When a node receives a packet, it identifies multiple candidate nodes within its transmission range.

3. Candidate Evaluation:-
→ Each candidate is evaluated based on factors like distance to the destination, remaining energy & link quality.

4. Path Selection:-
→ The node selects the most promising candidate based on the evaluation criteria.

5. Packet Forwarding:-
→ The packet is forwarded to the selected candidate node.

* Challenges:-

1. Routing voids:-
→ Geographic obstacles or sparse n/w may cause failures in greedy forwarding.

2. Localization Accuracy:-
→ The protocol heavily depends on accurate location data, which may not always be available or precise.

3. Load Balancing:-
→ Consistently selecting the closest node might lead to energy depletion in certain nodes, reducing n/w reliability.

* Applications:-

1. Wireless sensor networks (WSNs)
2. Mobile Ad hoc Networks (MANETs)
3. Internet of Things (IOT)
4. vehicular Ad Hoc Network (VANETS)

* Advantages:-

1. Reduced overhead
2. Scalability
3. Energy Conservation
4. Low latency
5. Robustness
6. Improved Path Quality.

TCP over Ad Hoc TCP Protocol Overview:-

→ Transmission Control Protocol is widely used for reliable Communication in traditional wired network.

→ However, implementing TCP in ASN Poses unique Challenges due to their dynamic topology, limited resources and shared wireless medium.

→ To address these, various adaptation and protocols have been developed for TCP over Ad hoc & sensor n/w.

* characteristics:-

1. Dynamic Topology:-

→ Nodes in ad hoc network frequently join, leave, or move, causing frequent route changes.

2. wireless Medium:-

→ shared and lossy wireless links lead to high Packet loss rates.

3. Energy Constraints:-

→ Sensor nodes often operate on limited Power, requiring energy-efficient Communication.

4. Bandwidth limitation:-
→ wireless links typically have lower bandwidth compared to wired networks.

* Challenges:-

1. Route Breaks and Packet loss:-
→ TCP interprets packet losses as Congestion.
→ Frequent route breaks lead to Performance degradation due to unnecessary retransmissions and reduced Congestion window sizes.

2. Hidden and Exposed node Problems.
→ Interference caused by hidden and exposed nodes leads to Packet collisions.

3. High overhead:-
→ TCP's acknowledgment - based mechanisms increase overhead in energy - Constrained sensor networks.

4. Multi-hop Communication:-
→ Multi-hop routing Exacerbates delays and Packet losses, further affecting TCP Performance.

5. Fairness issues:-
→ Competing TCP flows in shared wireless mediums often result in unfair bandwidth distribution.

* Enhancement to TCP for ASN / Key approaches:-

1. Split TCP
→ Splits the TCP Connection into multiple segments for better reliability and reduced end-to-end delay.
→ Each segment has its own acknowledgment system.

2. Explicit link failure Notification (ELFN)
→ Nodes notify the sender explicitly about route failures to avoid unnecessary Congestion Control actions.

3. TCP-F (TCP with Feedback)
→ Utilizes feedback from intermediate nodes to indicate Congestion or link failures, allowing the sender to take Corrective action.

4. Ad Hoc TCP (ATCP)
→ works in Conjunction with the n/w layer to detect n/w disconnections or Packet losses Caused by route failures and Congestion.

5. Rate-Based Protocols:-
→ Replace TCP's window-based congestion control with rate-based mechanisms to account for dynamic n/w conditions.

6. Cross-layer Approaches:-
→ share information across layers to improve TCP's adaptability to n/w conditions.

7. Energy-Aware TCP:-
→ Designed to minimize energy consumption by reducing redundant retransmissions and idle listening.

Examples:-

1. UDP:-
→ used for applications that can tolerate packet losses.

2. Reliable Data Protocols:-
→ Provide reliability with lower overhead.

3. Transport layer protocols for WSN:-
→ specifically designed for resource-constrained environments.

TCP and MANETS:-

→ Transmission Control Protocol is the dominant transport-layer protocol for reliable communication in traditional n/w.

→ However, applying TCP in Mobile Ad Hoc Networks (MANETs) and sensor n/w faces challenges due to the dynamic and resource-constrained nature of these n/w.

* Characteristics of MANETs in ASN:-

1. Dynamic Topology:-
→ Nodes frequently move, causing frequent changes in n/w topology.

2. Decentralization:-
→ No fixed infrastructure; nodes act as both hosts and routers.

3. Multi-Hop Communication:-
– Communication b/w nodes often involves multiple intermediate hops.

4. Energy Constraints:-
→ Mobile and sensor nodes are often battery-Powered
5. variable link Quality:-
→ wireless links are prone to interference, fading and high error rates.

* Issues with TCP in MANETs
→ TCP was originally designed for wired n/w with relatively stable topologies and low packet loss due to transmission errors.
→ In MANETs, TCP's assumptions lead to significant Performance Problems:

1. Misinterpreted Packet loss:-
→ In wired networks: Packet loss is usually due to congestion
→ In MANETs:- Packet loss often occurs due to link failures, mobility or interference.
→ TCP responds by reducing the congestion window unnecessarily, further degrading throughput.

2. Frequent Route changes:-
→ TCP struggles to maintain flow continuity due to frequent route updates caused by node mobility.

3. Hidden and Exposed Terminal Problems:-
→ Hidden nodes cause collisions, while Exposed nodes unnecessarily back off from transmissions, impacting throughput.

4. Network Partitioning:-
→ Node mobility can create temporary disconnections, leading to Packet loss and retransmission delays.

5. Unfair Bandwidth utilization.
→ TCP flows often compete unfairly in shared wireless environments, leading to throughput imbalances.

6. High overheads:-
→ TCP's Acknowledgment mechanism and retransmissions increase the overhead, which is inefficient in energy constrained sensor n/w.

* Alternatives to TCP in MANETS.

1. UDP (User Datagram protocol):-
→ lightweight and suitable for applications that can tolerate some Packet loss.

2. Transport Protocols for MANETs:-
→ AODV-TCP: Combines AODV routing with TCP enhancements.
→ TCP-Bus (Backup source Routing):- Recovers quickly from route failures by maintaining backup routes.

3. Protocols for Sensor Networks:-
→ PSFQ (Pump slowly, fetch quickly):- Ensures reliability with minimal overhead.
→ CODA (Congestion Detection and Avoidance): Reduce Congestion by Controlling traffic Sources.

Solutions for TCP Over Ad-Hoc:-

→ The unique Characteristics of ASN, such as dynamic topology, limited bandwidth, energy constraints and multi-hop Communication, make traditional TCP inefficient.
→ To address these issues, several solutions and optimizations have been proposed.
→ These Solutions focus on adapting TCP's mechanisms to the challenges of ASN.

1. Explicit link failure Notification (ELFN)
→ Problem Addressed:- TCP misinterprets route failures as Congestion.
→ solution:-
→ intermediate nodes or the n/w layer send explicit notifications to the TCP Sender when a link or route fails.
→ upon receiving the ELFN, the sender pauses its timers and stops reducing the Congestion window unnecessarily.

→ Transmission resumes once the route is restored.

2. Feedback-Based TCP (TCP-F):-

→ Problem Addressed:-

· lack of distinction b/w Congestion losses and mobility-induced losses.

→ solution:-

→ Nodes in the n/w Send feedback to the sender about the state of the n/w.

→ The sender adjusts its behavior accordingly, avoiding unnecessary retransmissions and Congestion Control actions.

3. Split-TCP:-

→ Problem Addressed:-

long end-to-end delays in multi-hop Communication

→ Solution:-

→ Divide the TCP Connection into multiple segments.

→ Each segment manages flow Control and error recovery independently, reducing latency and improving throughput.

4. Ad Hoc TCP (ATCP):-

→ Problem Addressed:-

TCP inability to distinguish b/w n/w disconnections, Congestion and link failures

→ solutions:-

→ ATCP operates as a thin layer b/w TCP and IP

→ Monitors n/w status using ICMP or other mechanisms.

→ Adjusts TCP behavior dynamically based on the n/w Conditions.

→ Pauses TCP during disconnections.

→ Reduce the Congestion window during Congestion.

5. TCP-Bus (TCP with Backup Source Routing):-

→ Problem Addressed:-

Frequent route failures due to mobility.

→ Solution:-

→ utilizes backup source routes to recover quickly from route failures without triggering TCP's Congestion Control mechanisms.

→ Reduces Packet loss and retransmissions caused by mobility.

6. Energy-Aware TCP:-

→ Problem Addressed:-

High energy consumption in sensor n/w.

→ Solution:-

→ Minimizes retransmissions by optimizing congestion control mechanisms.

→ Reduces the acknowledgment overhead by using cumulative or selective Acks.

7. Cross-layer optimizations:-

→ Problem Addressed:-

lack of coordination b/w layers in traditional TCP.

→ Solution:-

→ Enable information sharing b/w layers.

8. Multipath Routing and TCP:-

→ Problem Addressed:-

Single-path routing limits robustness and throughput.

→ Solution:-

→ use multipath routing to distribute TCP traffic across multiple paths.

→ Increases reliability and load balancing while mitigating the impact of route failures.